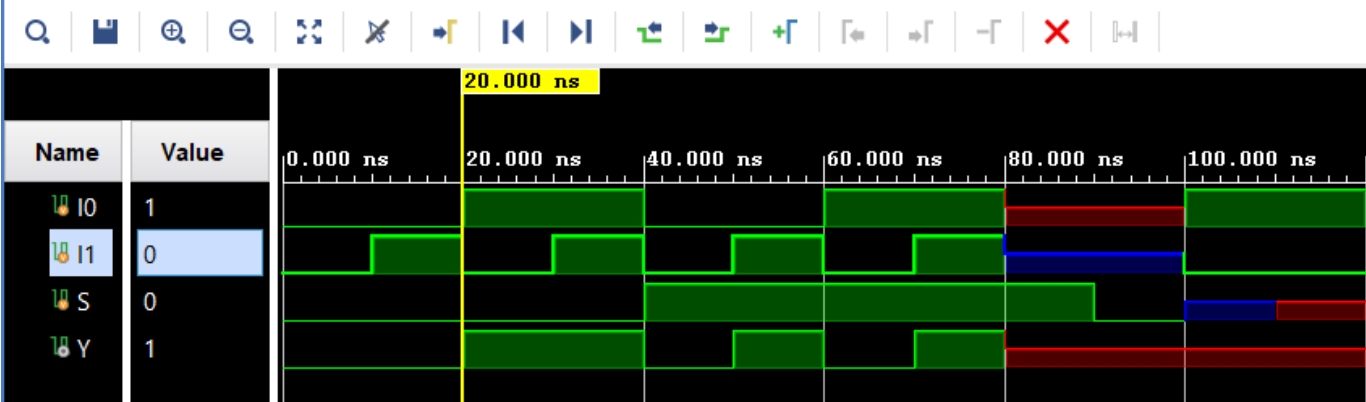
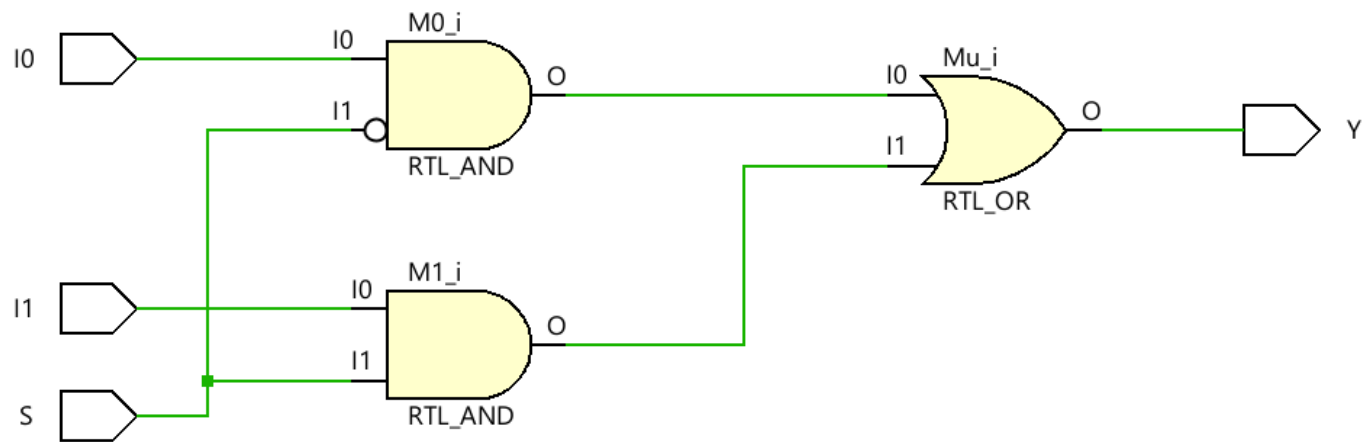


Multiplexer

Mux_2X

Structural Model

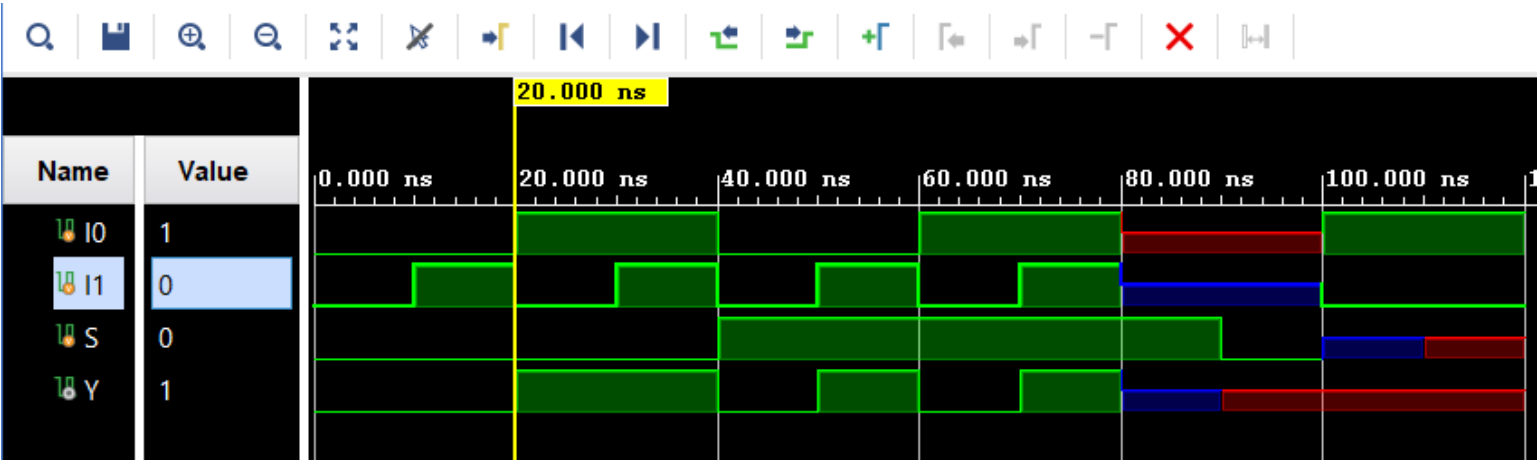
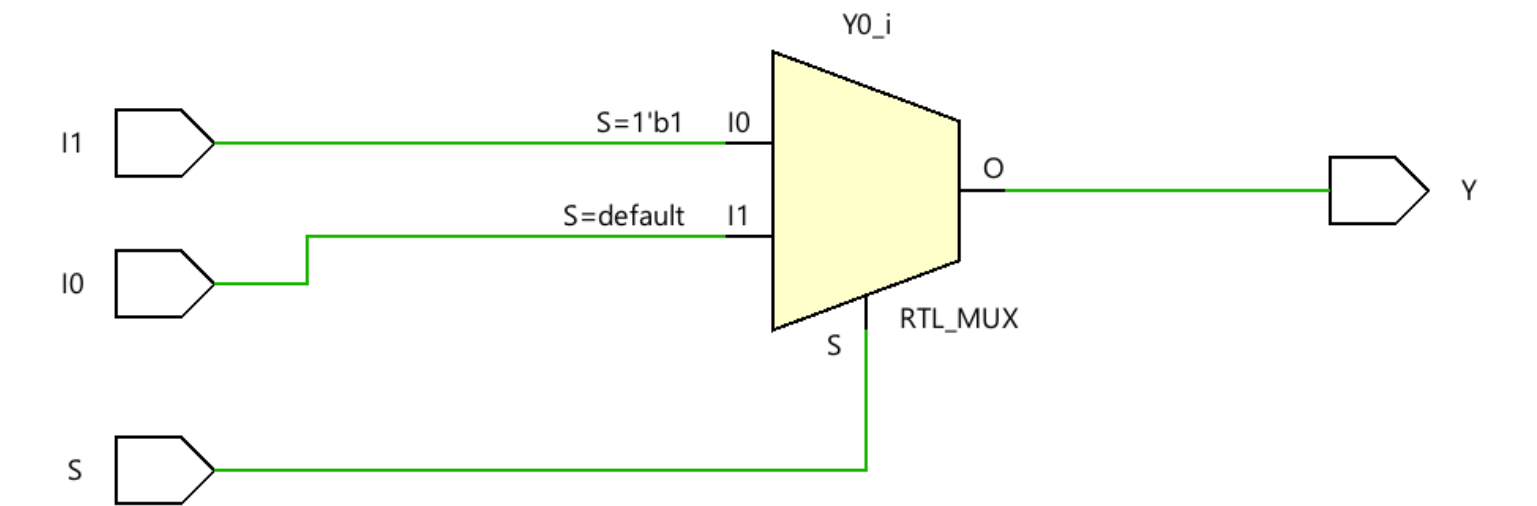


I0 | I1 | S | Y

0 | 0 | 0 | 0
0 | 1 | 0 | 0
1 | 0 | 0 | 1
1 | 1 | 0 | 1
0 | 0 | 1 | 0
0 | 1 | 1 | 1
1 | 0 | 1 | 0
1 | 1 | 1 | 1

x | z | 1 | x
x | z | 0 | x
1 | 0 | z | x
1 | 0 | x | x

Behavioral Modeling



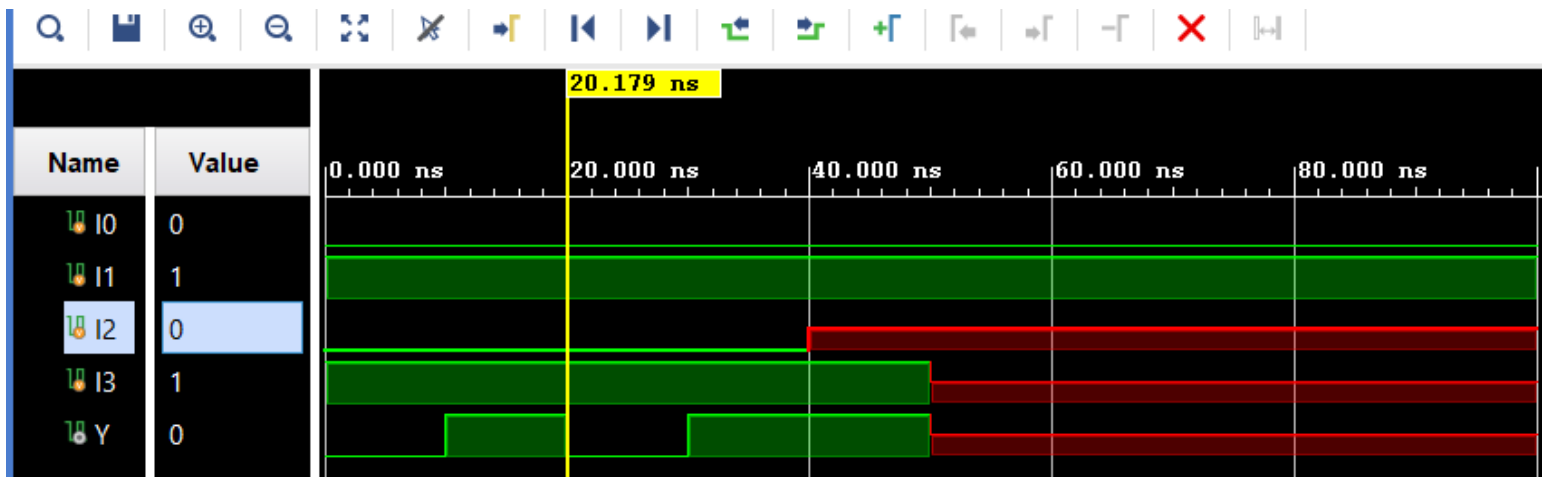
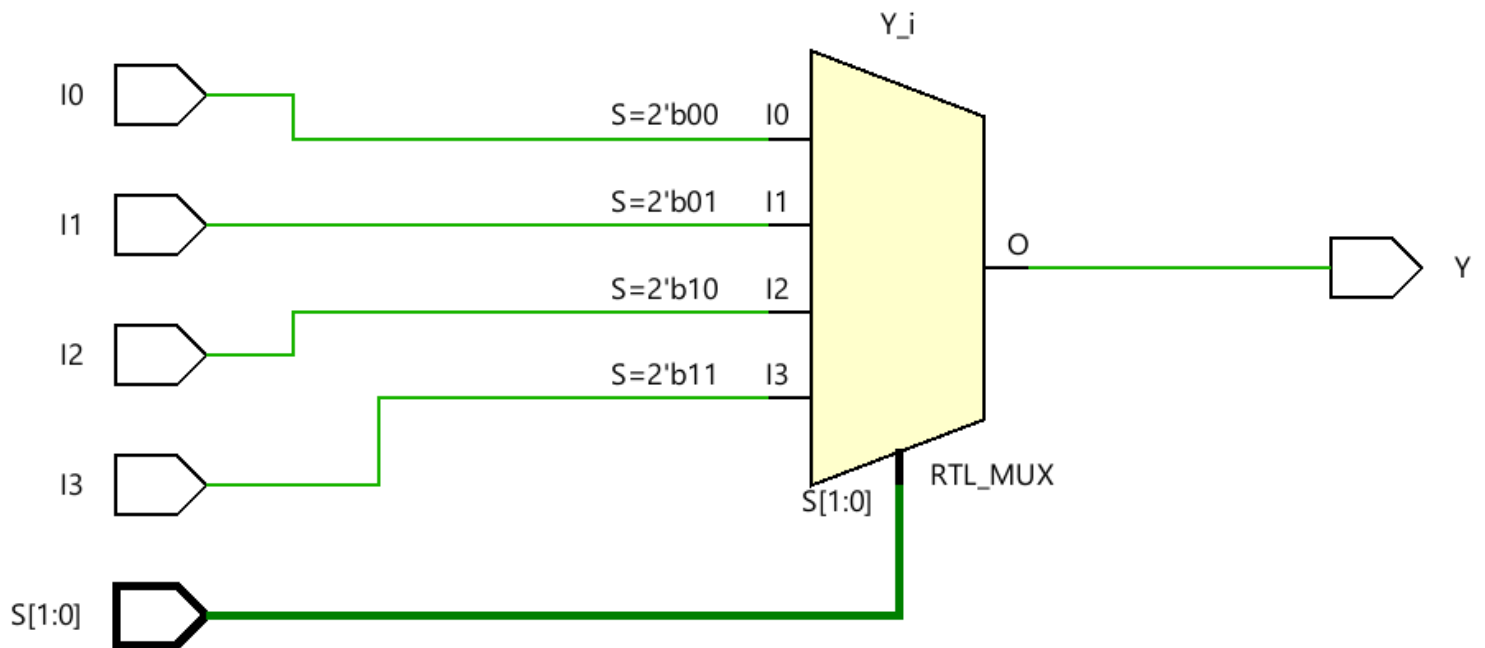
I0	I1	S	Y

0	0	0	0
0	1	0	0
1	0	0	1
1	1	0	1
0	0	1	0
0	1	1	1
1	0	1	0
1	1	1	1

x	z	1	z
x	z	0	x
1	0	z	x
1	0	x	x

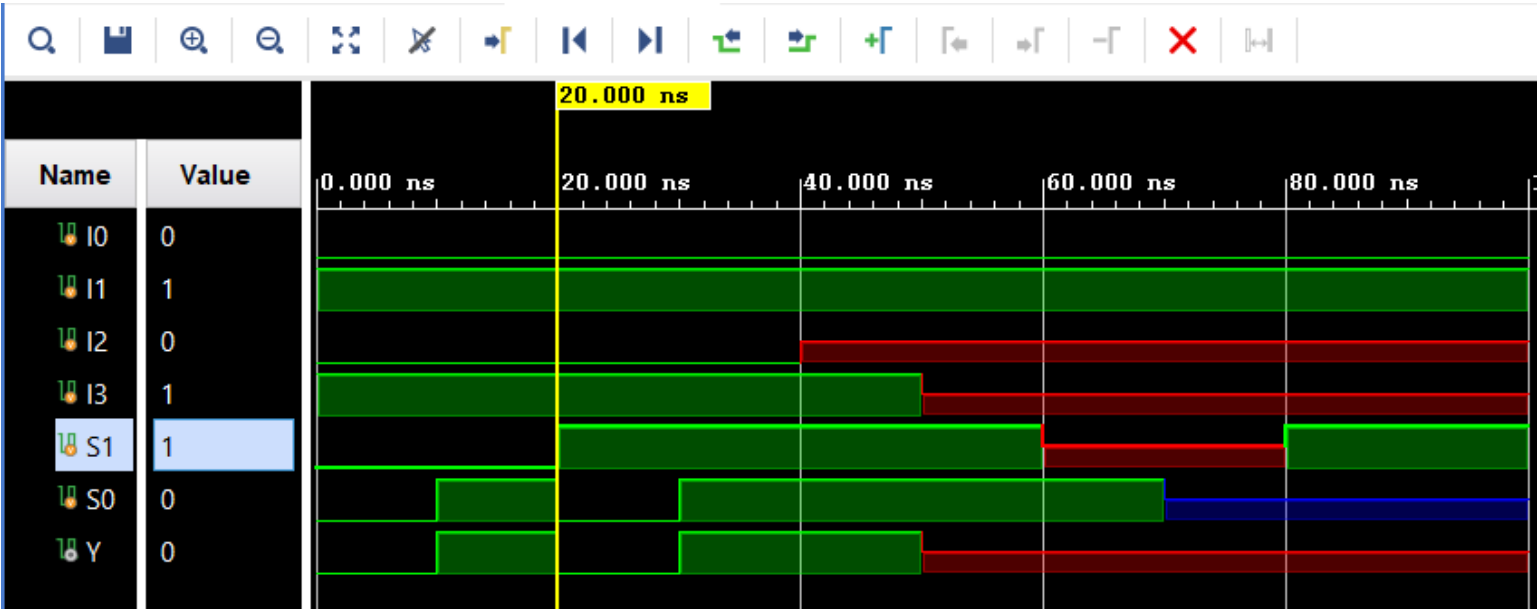
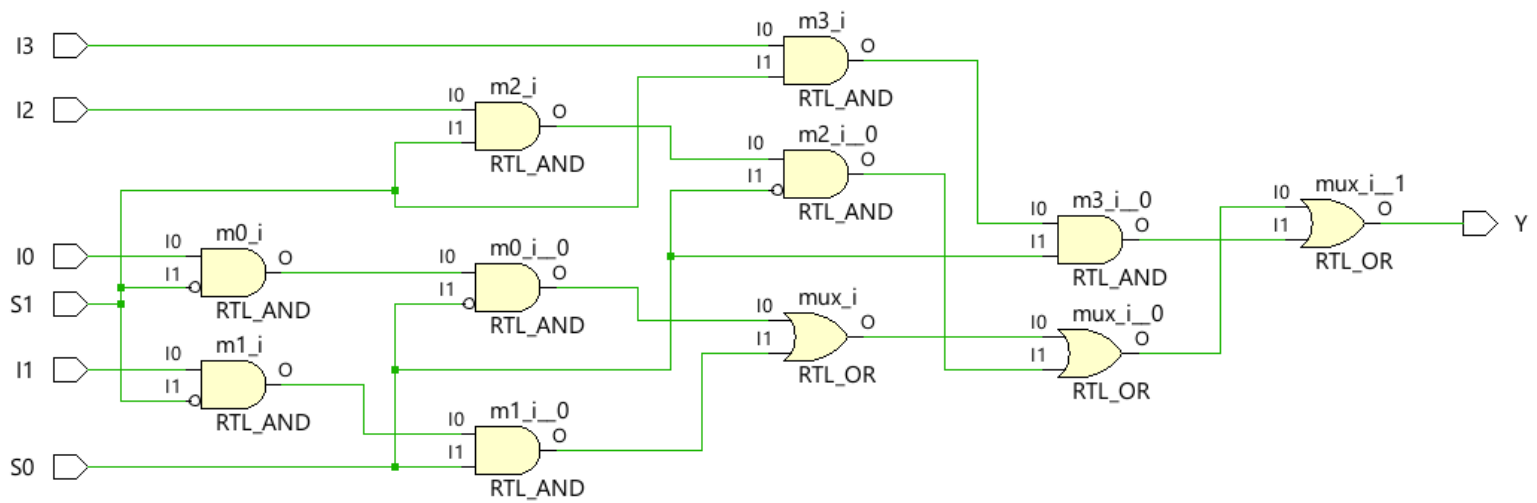
Mux_4X1

Behavioral Modeling



I	S1	S0	Y
0101	0	0	0
0101	0	1	1
0101	1	0	0
0101	1	1	1
01x1	1	1	1
01xx	1	1	x
01xx	x	1	x
01xx	x	z	x
01xx	1	z	x

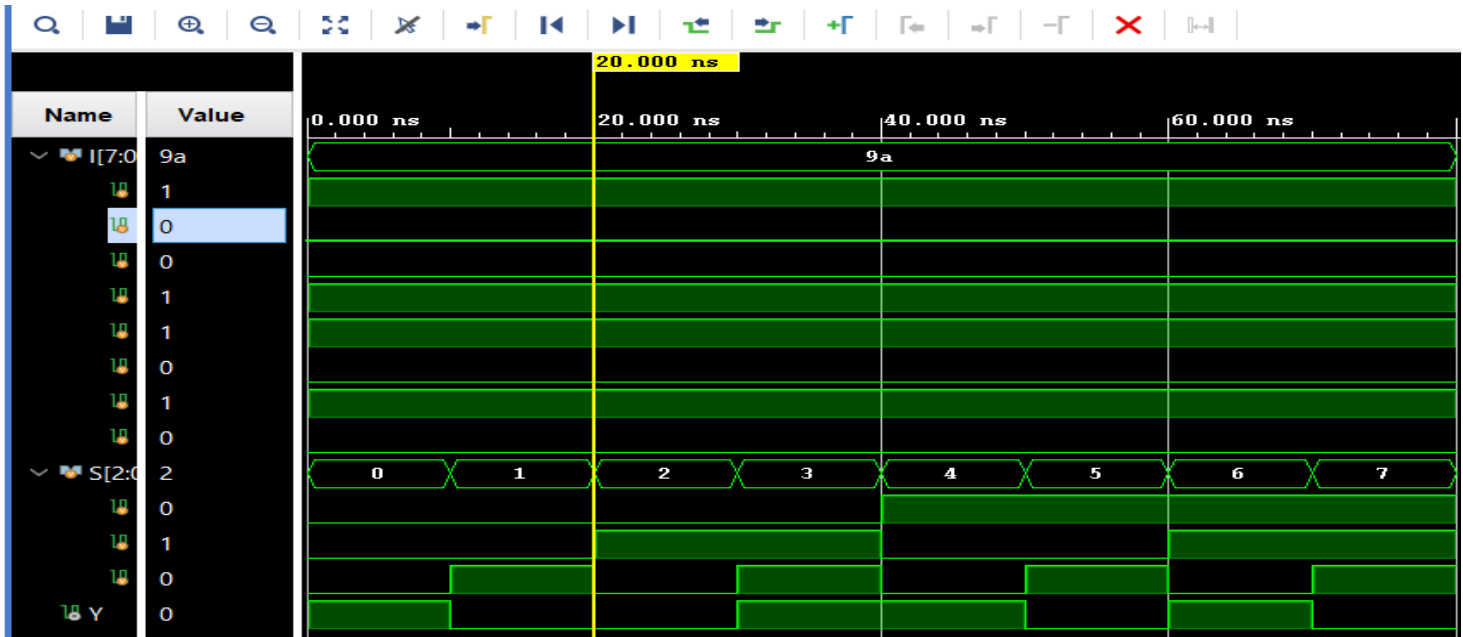
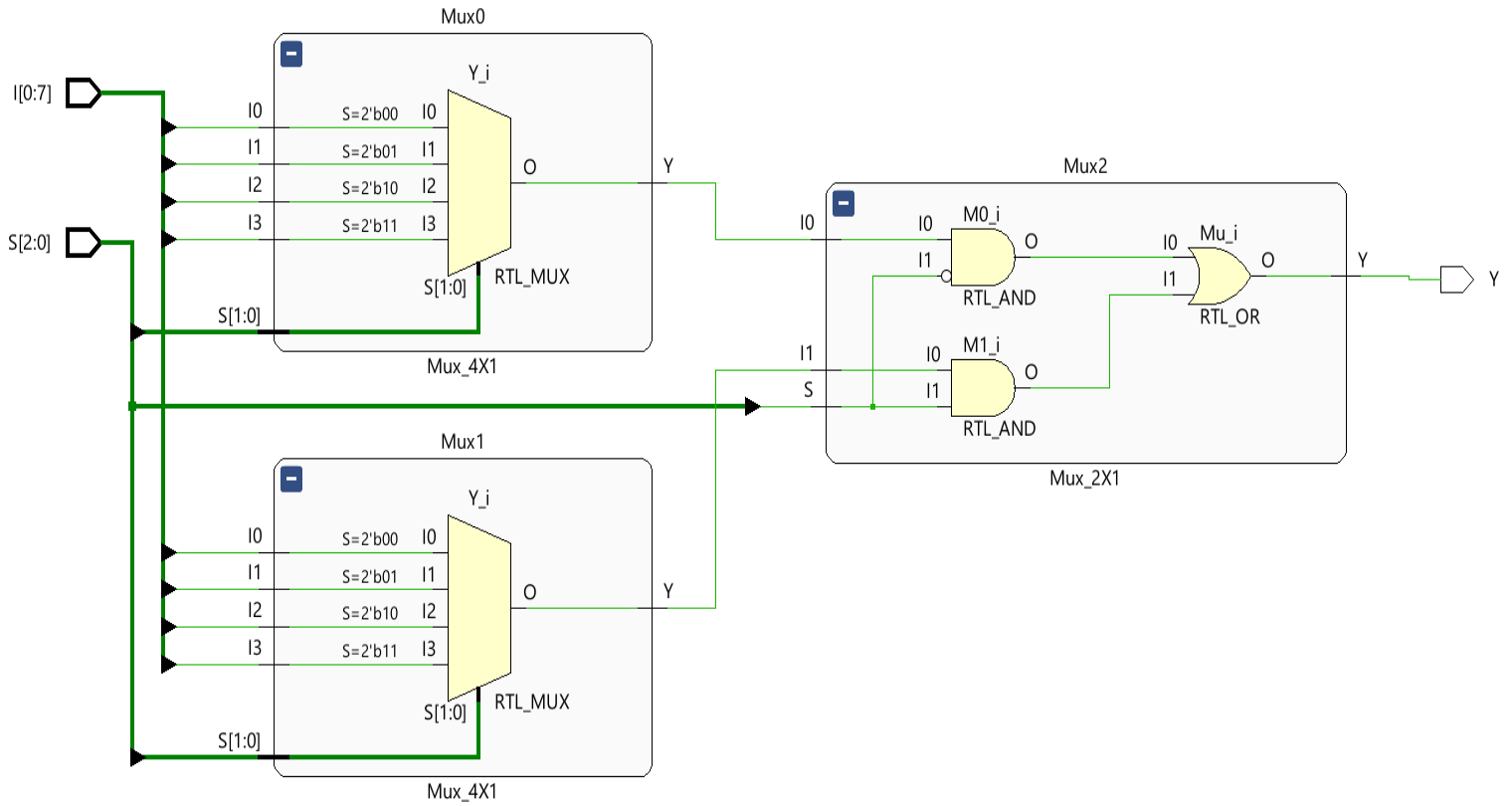
Structural Modeling



I	S1	S0	Y
0101	0	0	0
0101	0	1	1
0101	1	0	0
0101	1	1	1
01x1	1	1	1
01xx	1	1	x
01xx	x	1	x
01xx	x	z	x
01xx	1	z	x

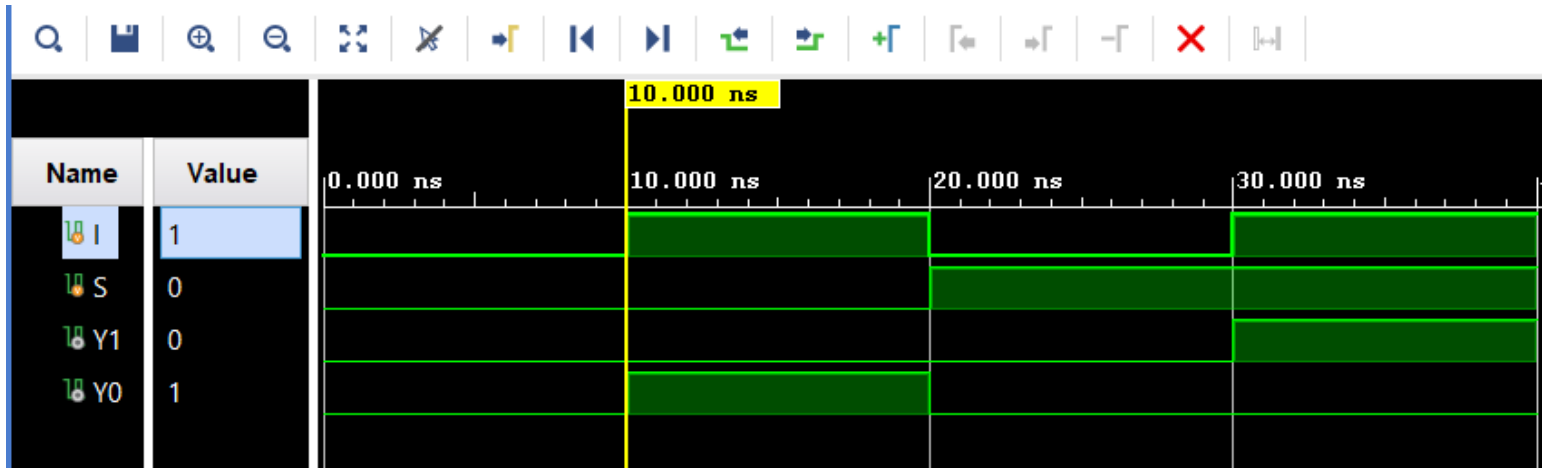
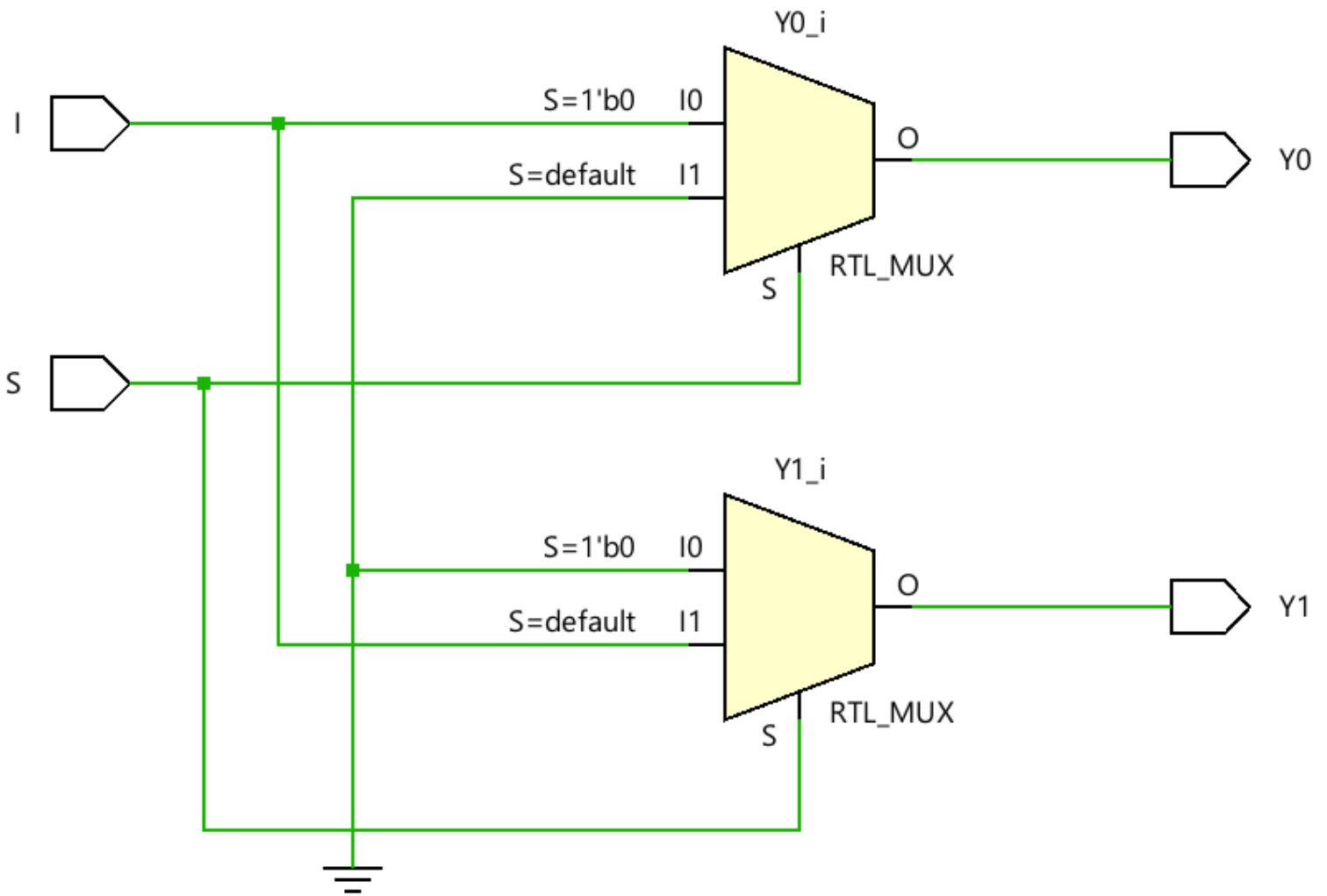
Mux_8X1

Structural Modeling



Time (ns)	I[7:0]	S[2:0]	Y
0	I=10011010	S=000	Y=1
10	I=10011010	S=001	Y=0
20	I=10011010	S=010	Y=0
30	I=10011010	S=011	Y=1
40	I=10011010	S=100	Y=1
50	I=10011010	S=101	Y=0
60	I=10011010	S=110	Y=1
70	I=10011010	S=111	Y=0

Demultiplexer

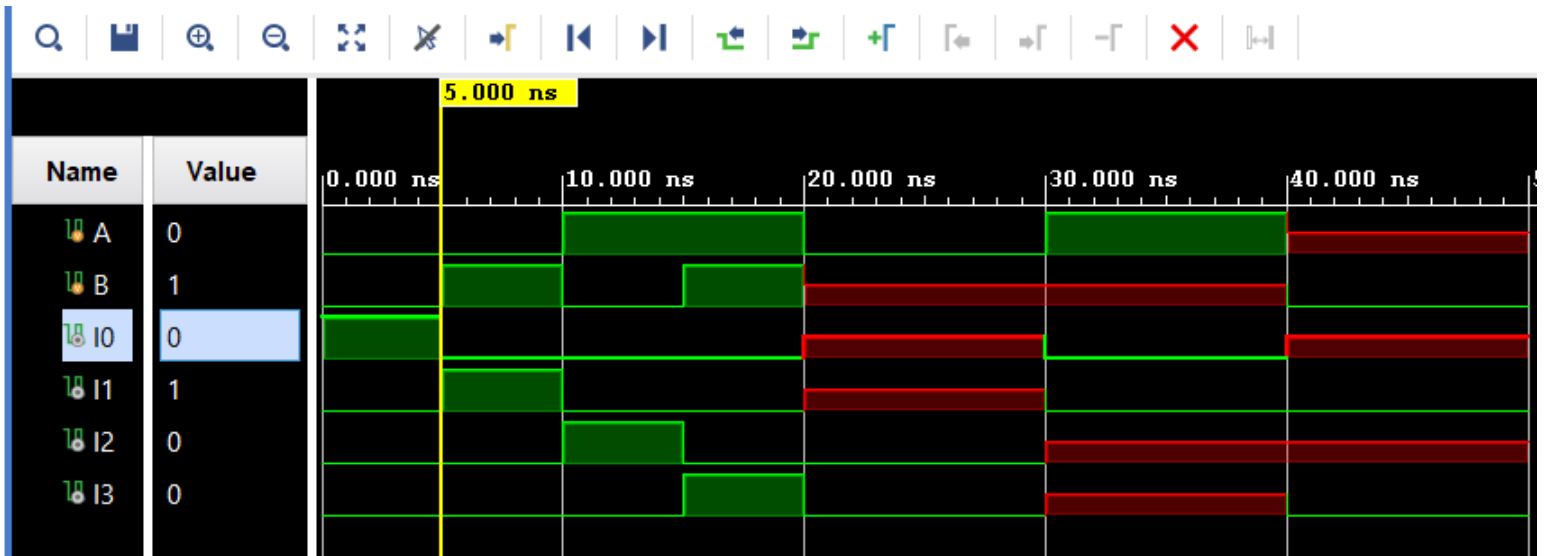
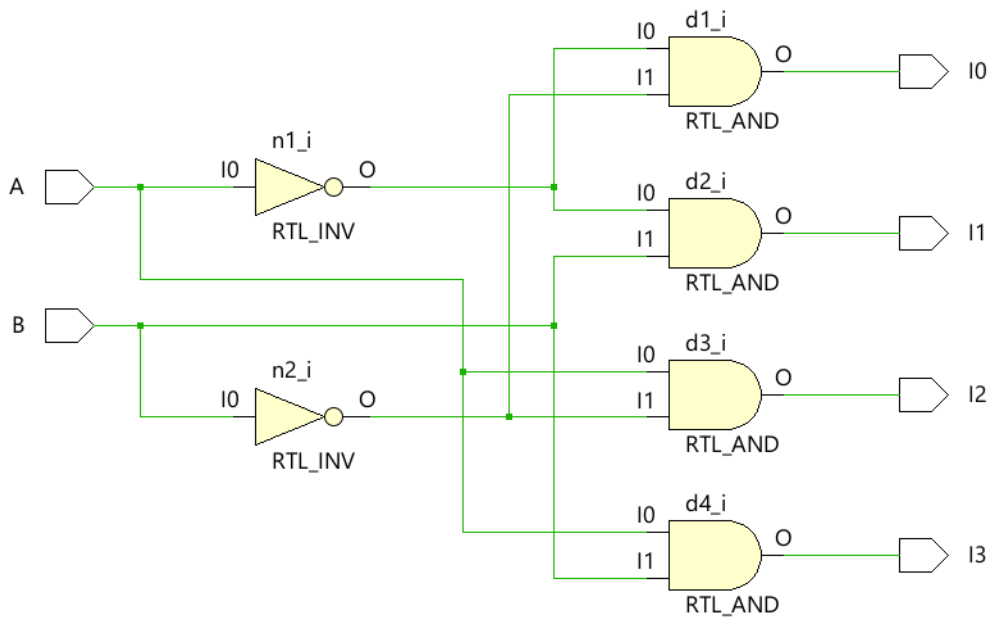


time	I	S	Y1	Y0
0	0	0	0	0
10	1	0	0	1
20	0	1	0	0
30	1	1	1	0

Decoder

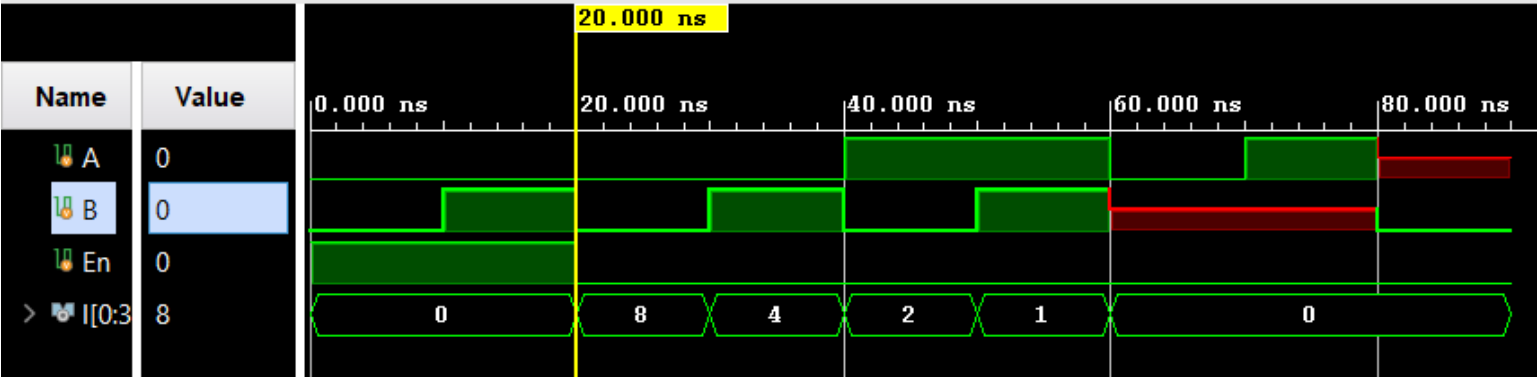
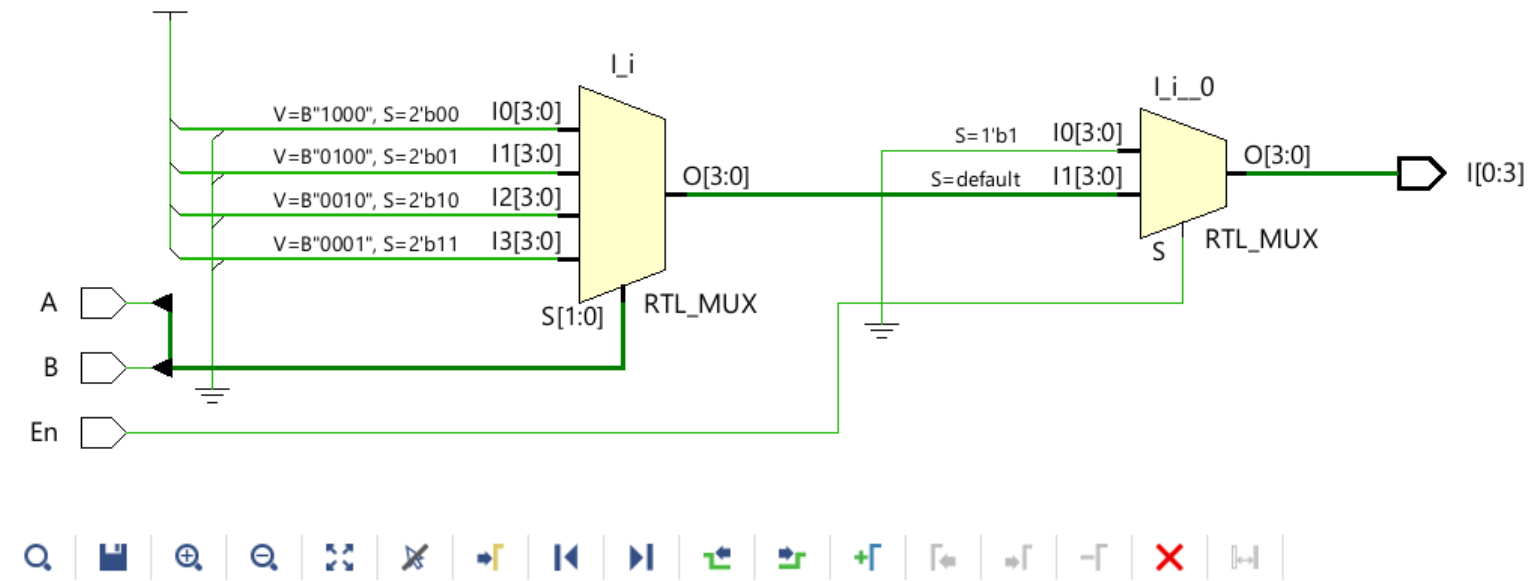
Decoder_2X4

Structural



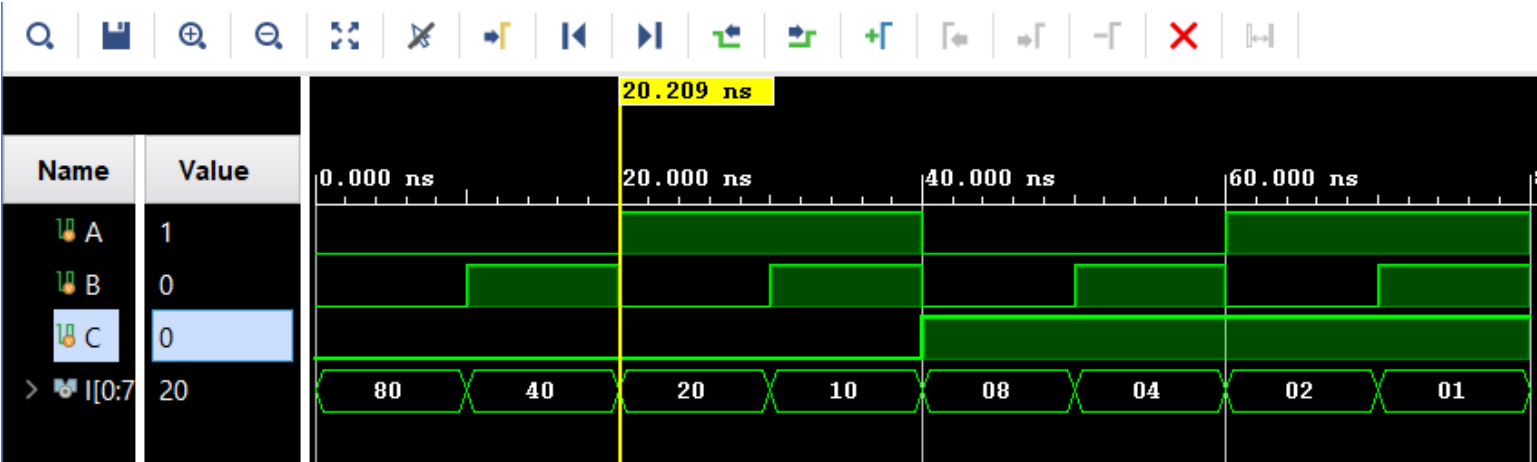
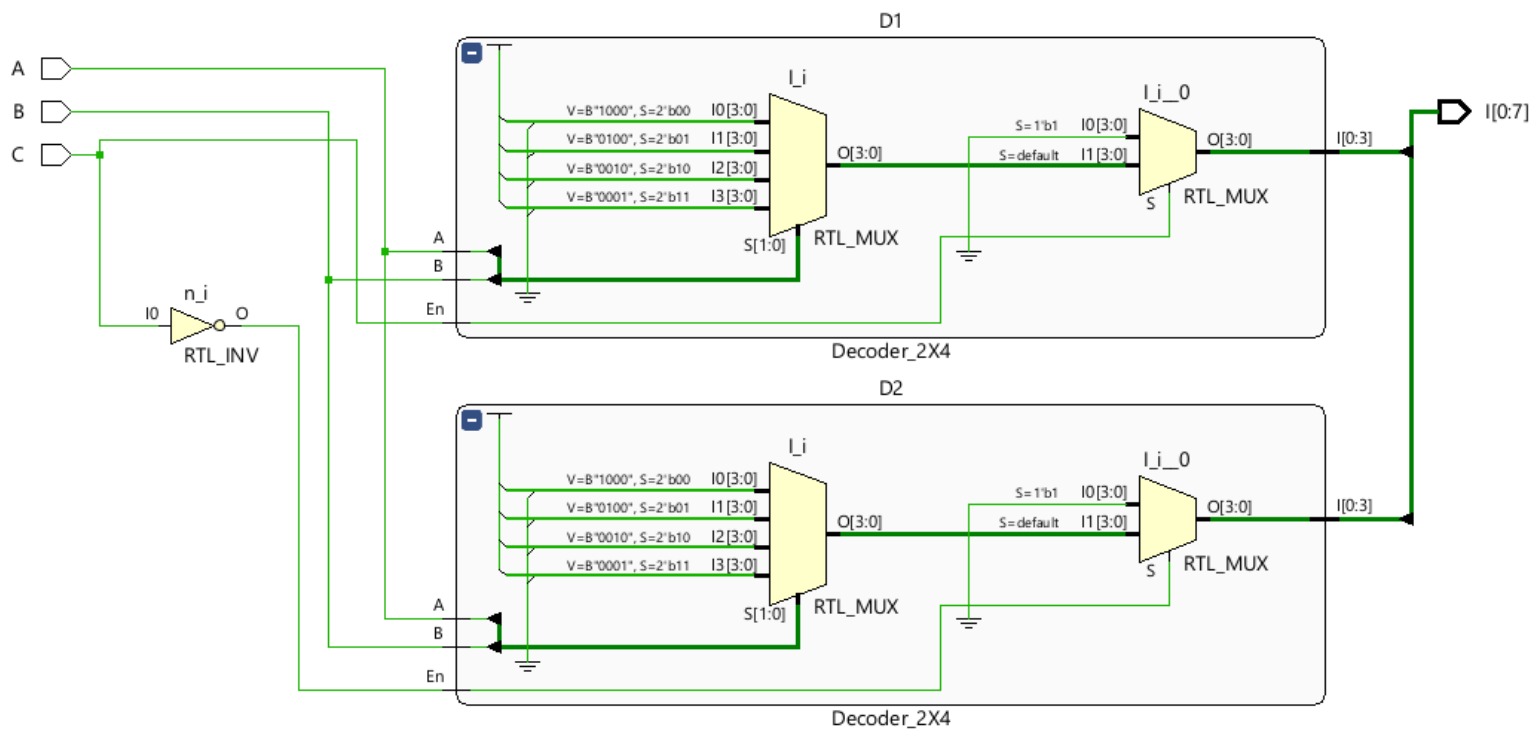
A	B	I0	I1	I2	I3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1
0	x	x	x	0	0
1	x	0	0	x	x
x	0	x	0	x	0

Behavioral



EN	A	B	I
1	0	0	0000
1	0	1	0000
0	0	0	1000
0	0	1	0100
0	1	0	0010
0	1	1	0001
0	0	x	0000
0	1	x	0000
0	x	0	0000

Decoder_3X8

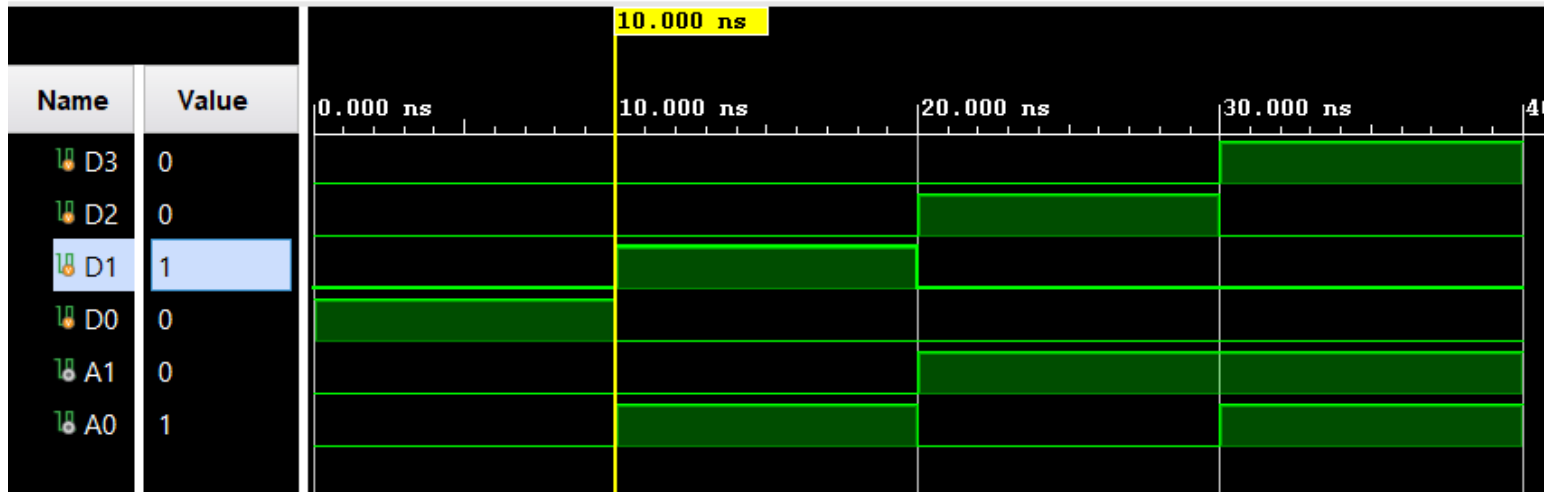
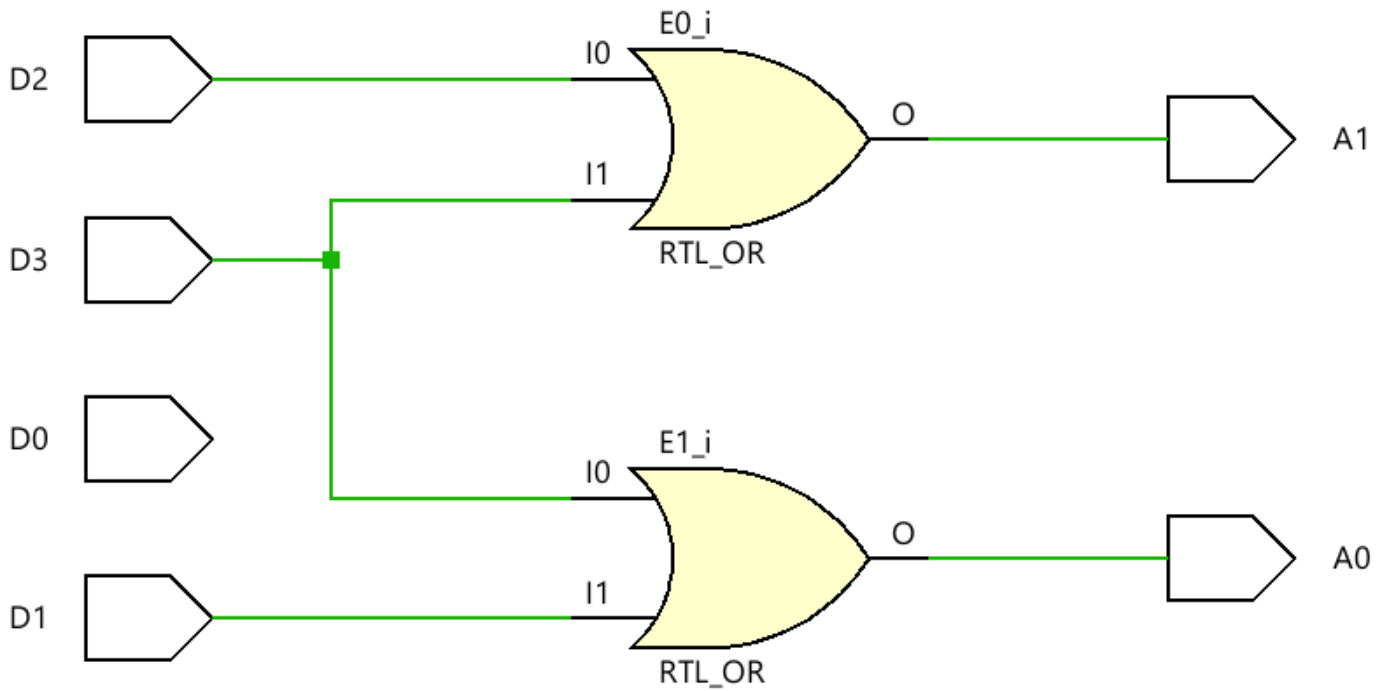


C | A | B || I[0:7]

0	0	0		10000000
0	0	1		01000000
0	1	0		00100000
0	1	1		00010000
1	0	0		00001000
1	0	1		00000100
1	1	0		00000010
1	1	1		00000001

Encoder

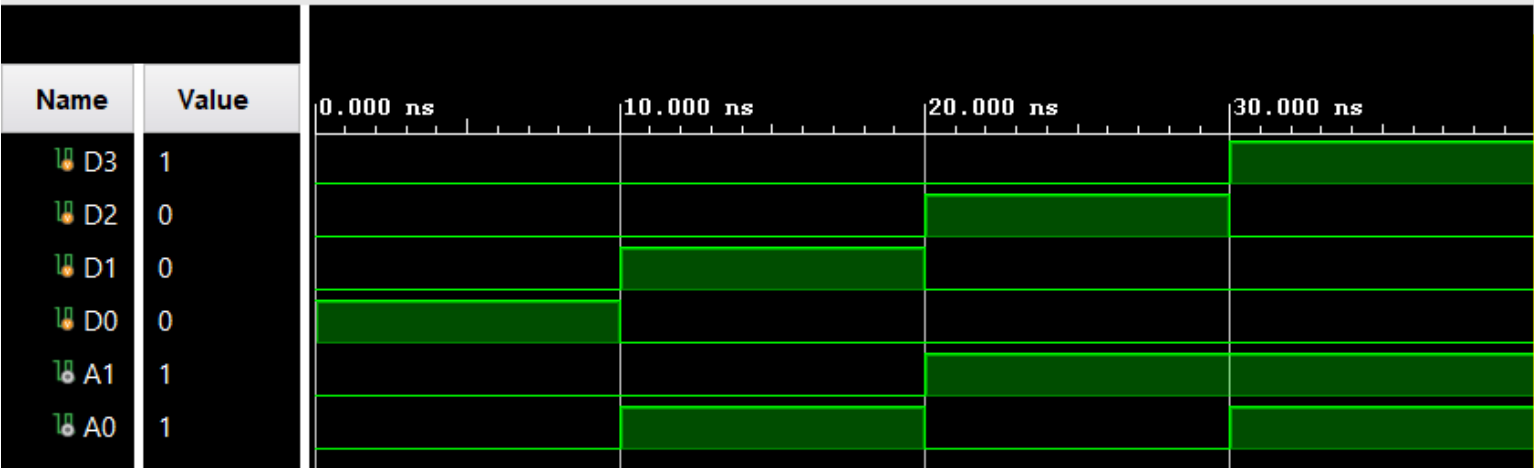
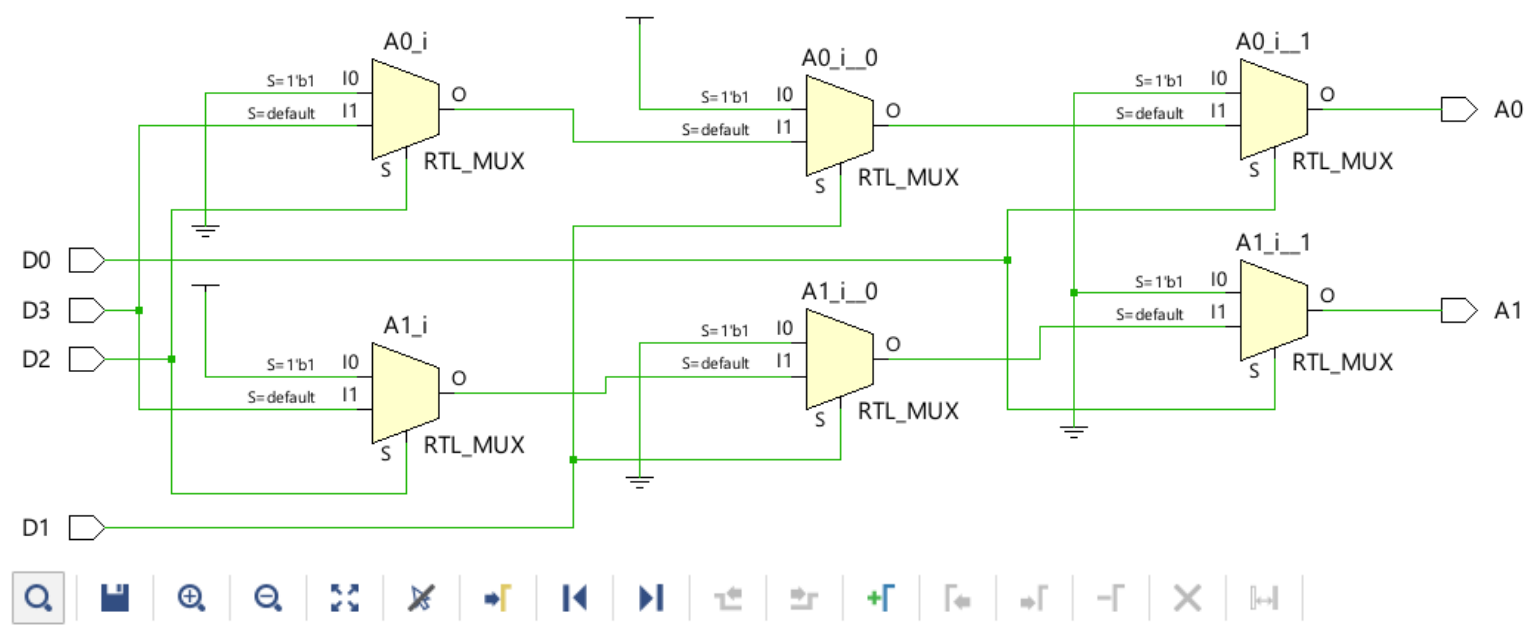
Encoder_4X2



D3 | D2 | D1 | D0 | | A1 | A0

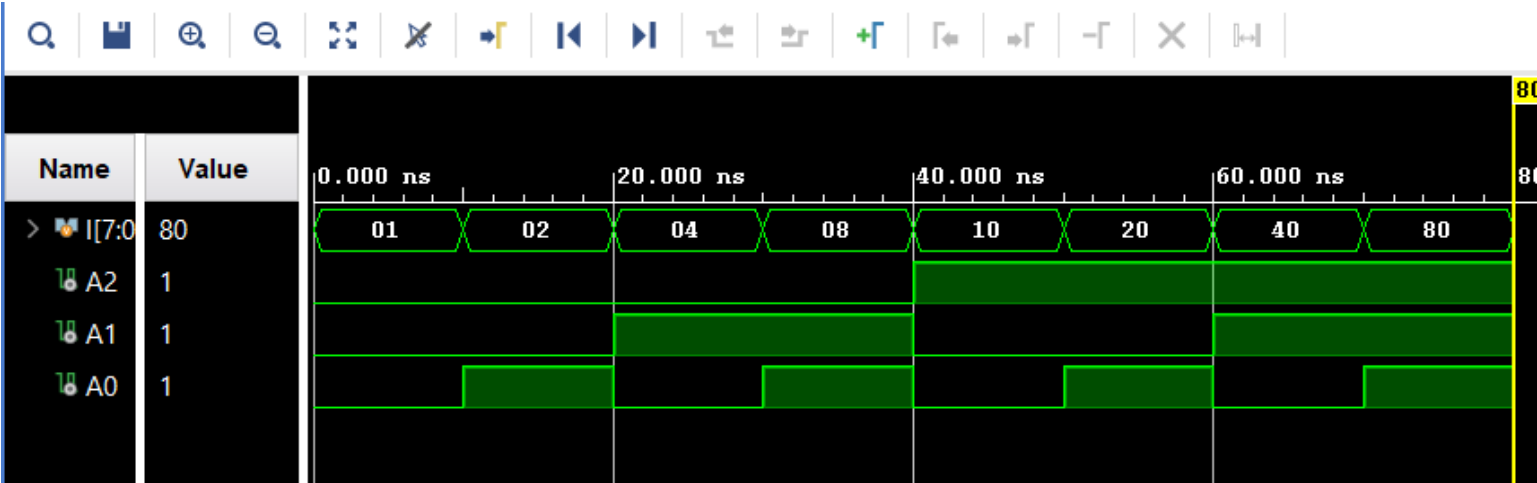
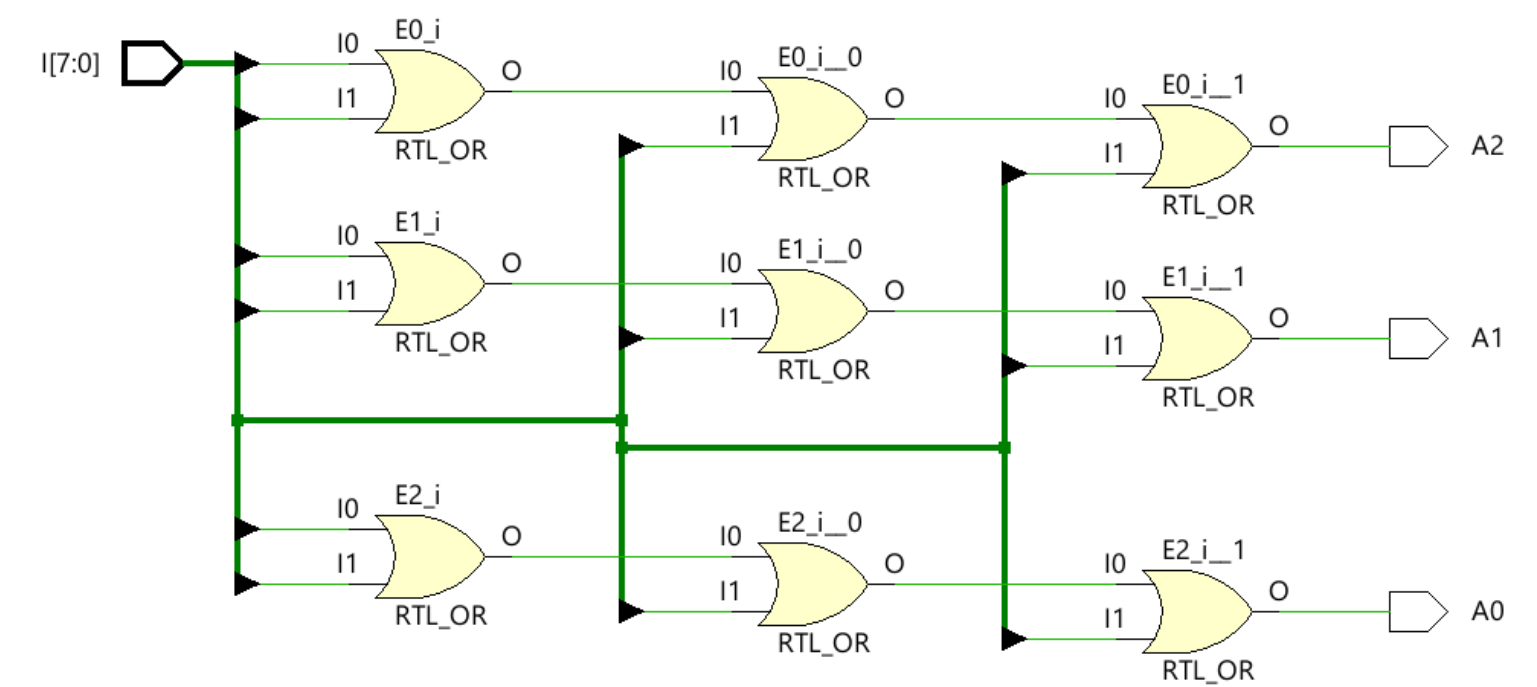
0	0	0	1		0	0
0	0	1	0		0	1
0	1	0	0		1	0
1	0	0	0		1	1

Behavioral



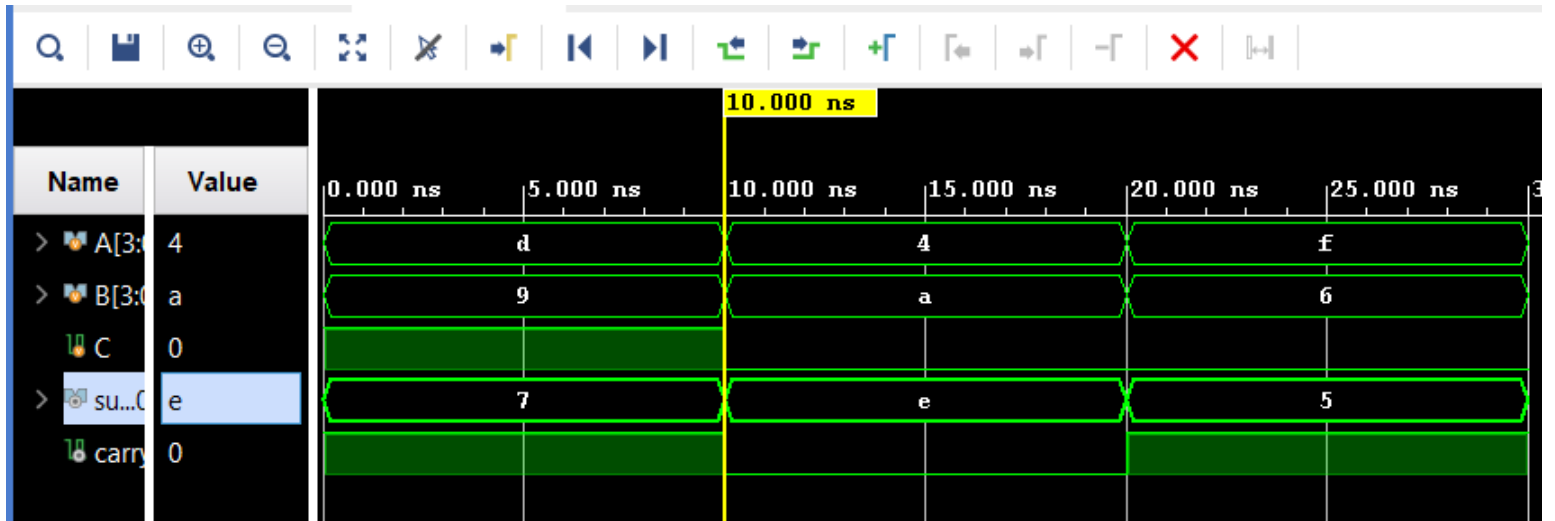
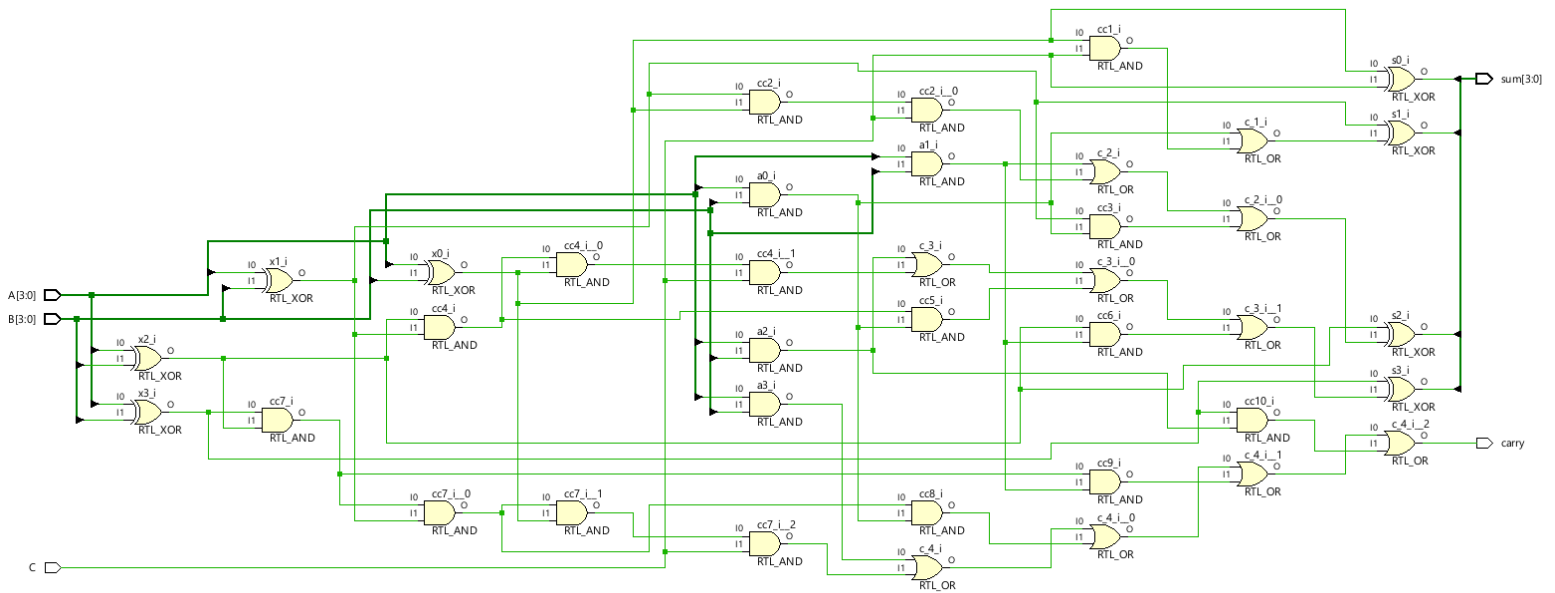
D3 D2 D1 D0 A1 A0							
0	0	0	1		0	0	
0	0	1	0		0	1	
0	1	0	0		1	0	
1	0	0	0		1	1	

Encoder_8X3



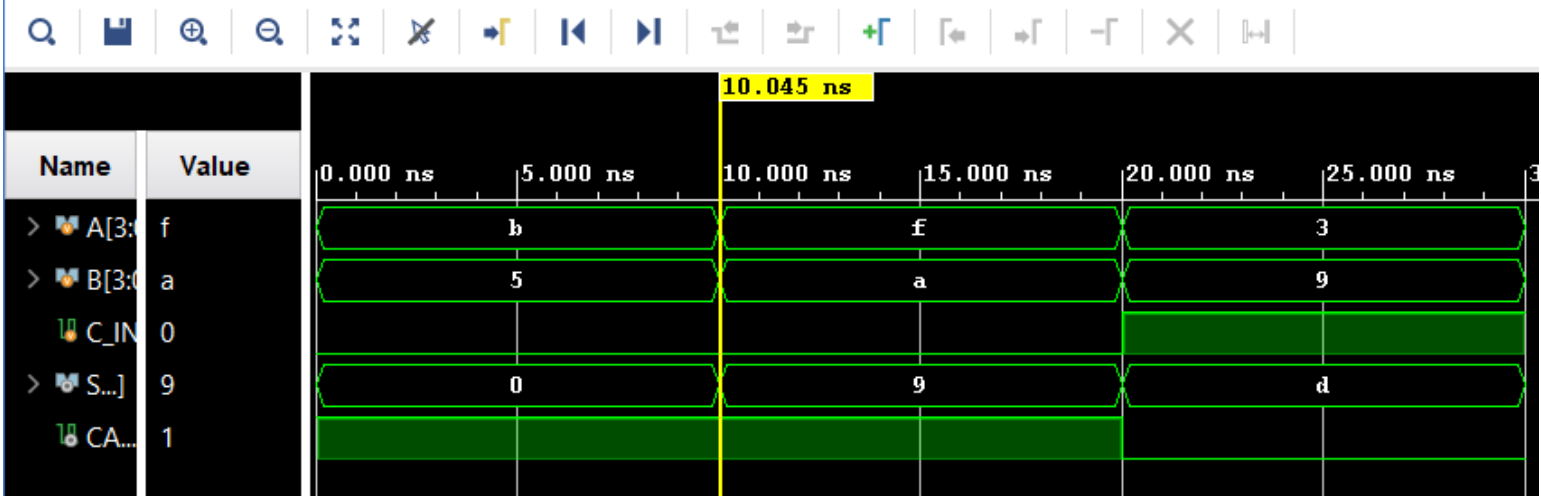
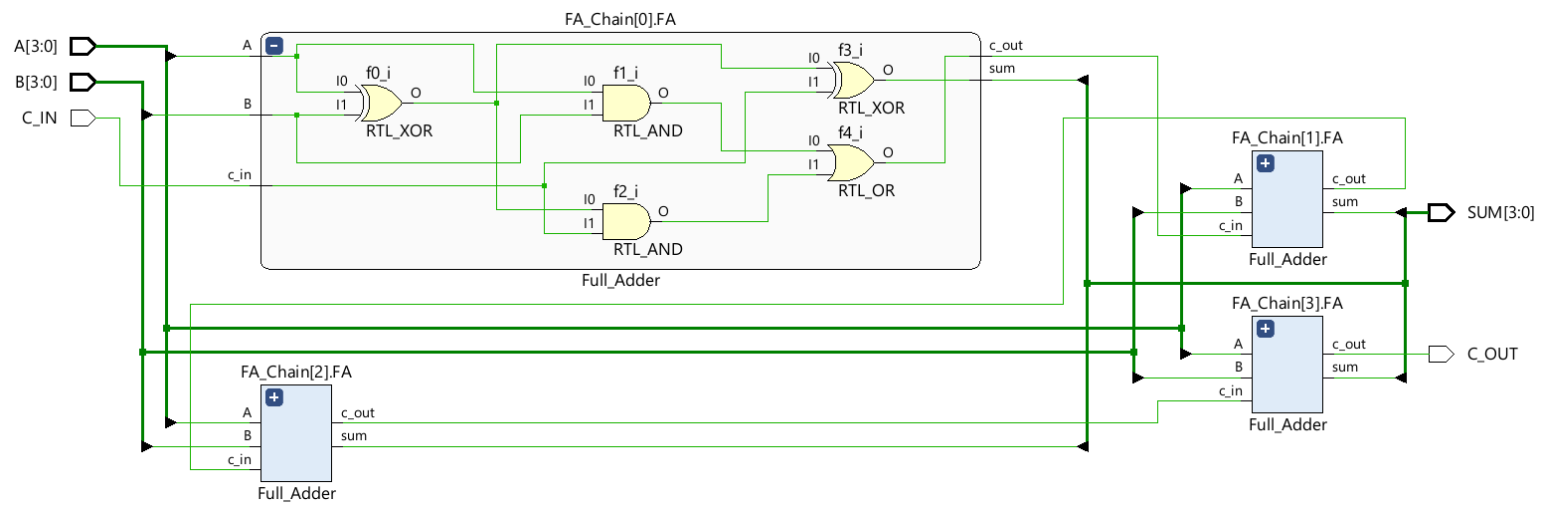
I A2 A1 A0				
00000001		0	0	0
00000010		0	0	1
00000100		0	1	0
00001000		0	1	1
00010000		1	0	0
00100000		1	0	1
01000000		1	1	0
10000000		1	1	1

Cla



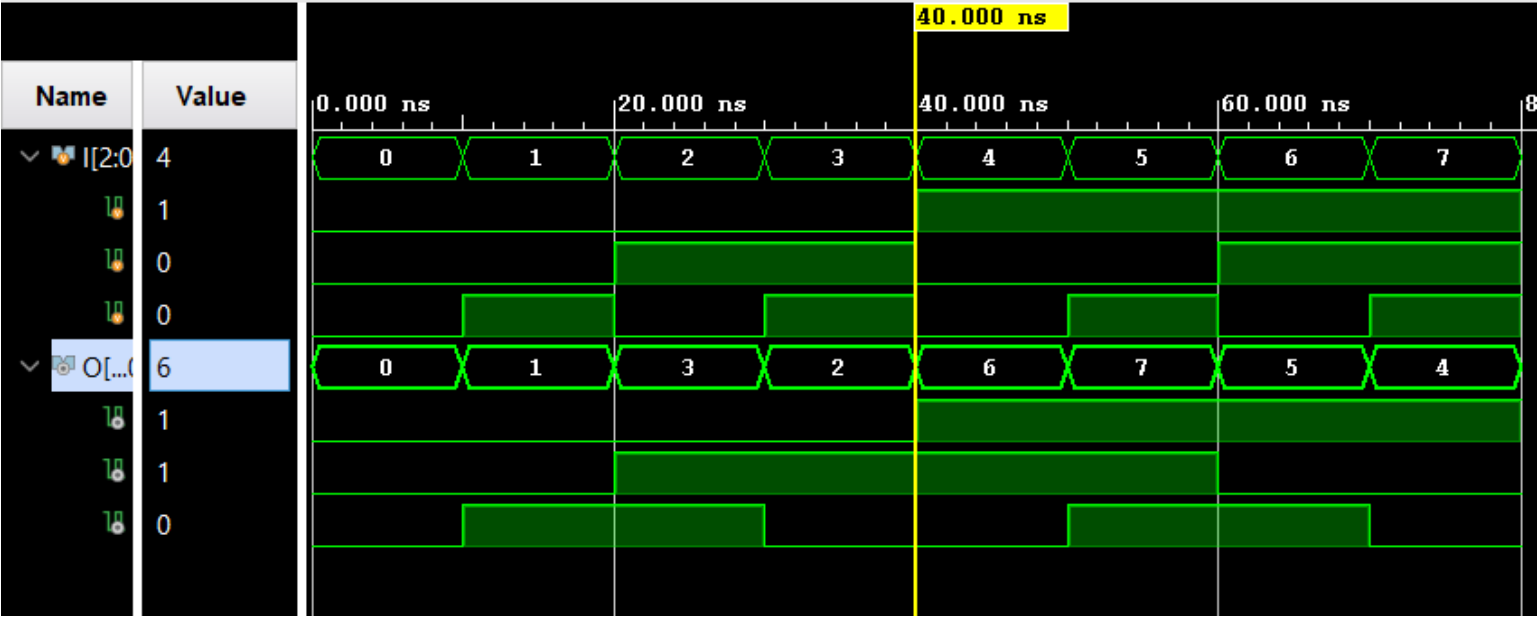
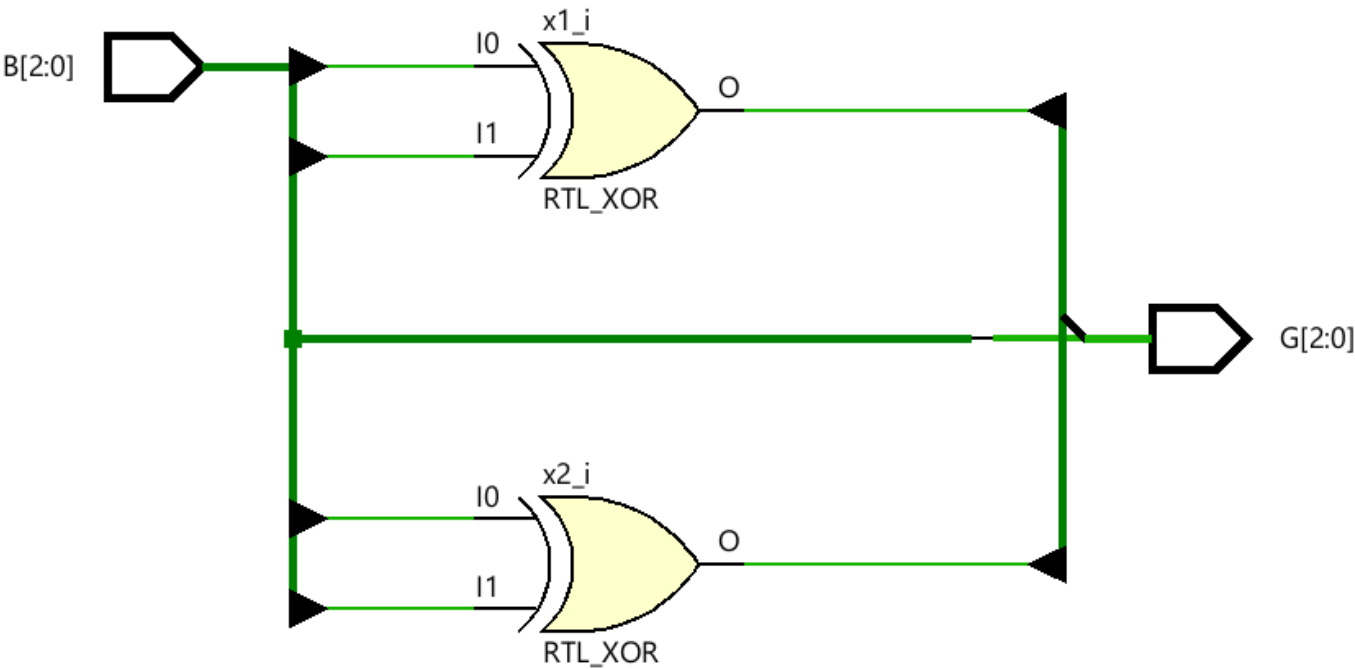
A	B	C	Sum	Carry
1101	1001	1	0111	1
0100	1010	0	1110	0
1111	0110	0	0101	1

N Bit Adder



0 A = 1011, B = 0101, Sum = 0000 , Carry = 1
10 A = 1111, B = 1010, Sum = 1001 , Carry = 1
20 A = 0011, B = 1001, Sum = 1101 , Carry = 0

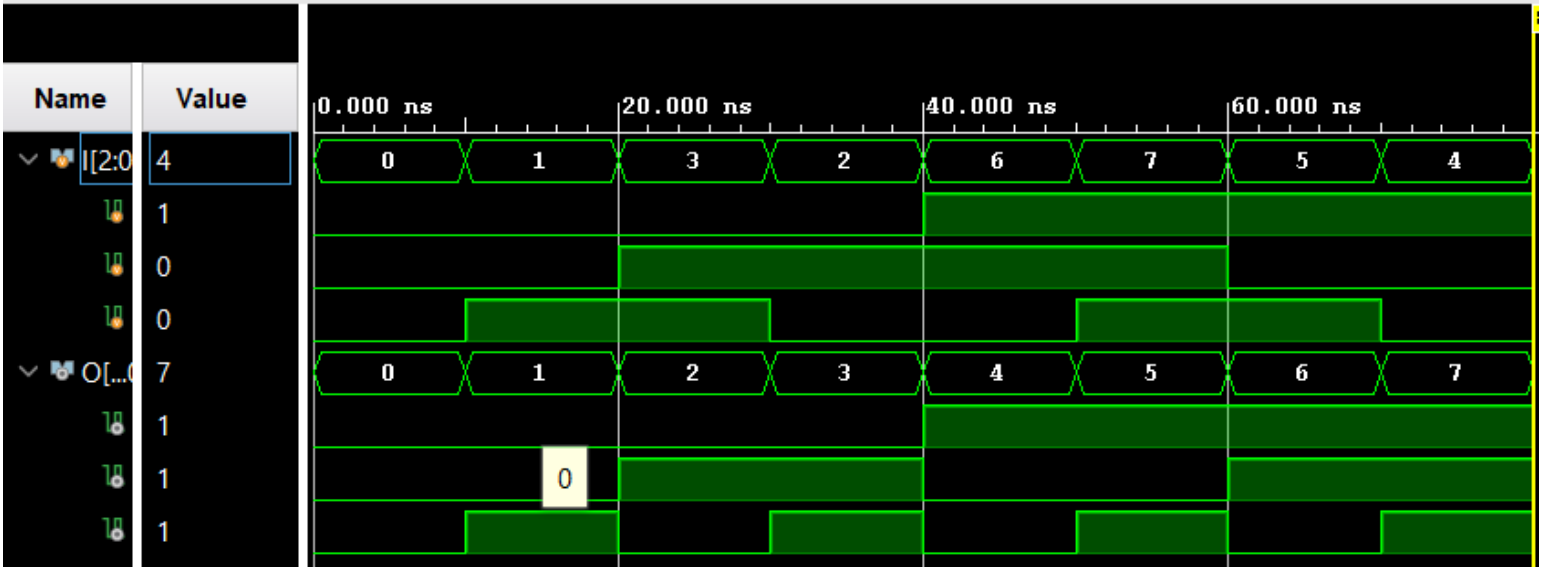
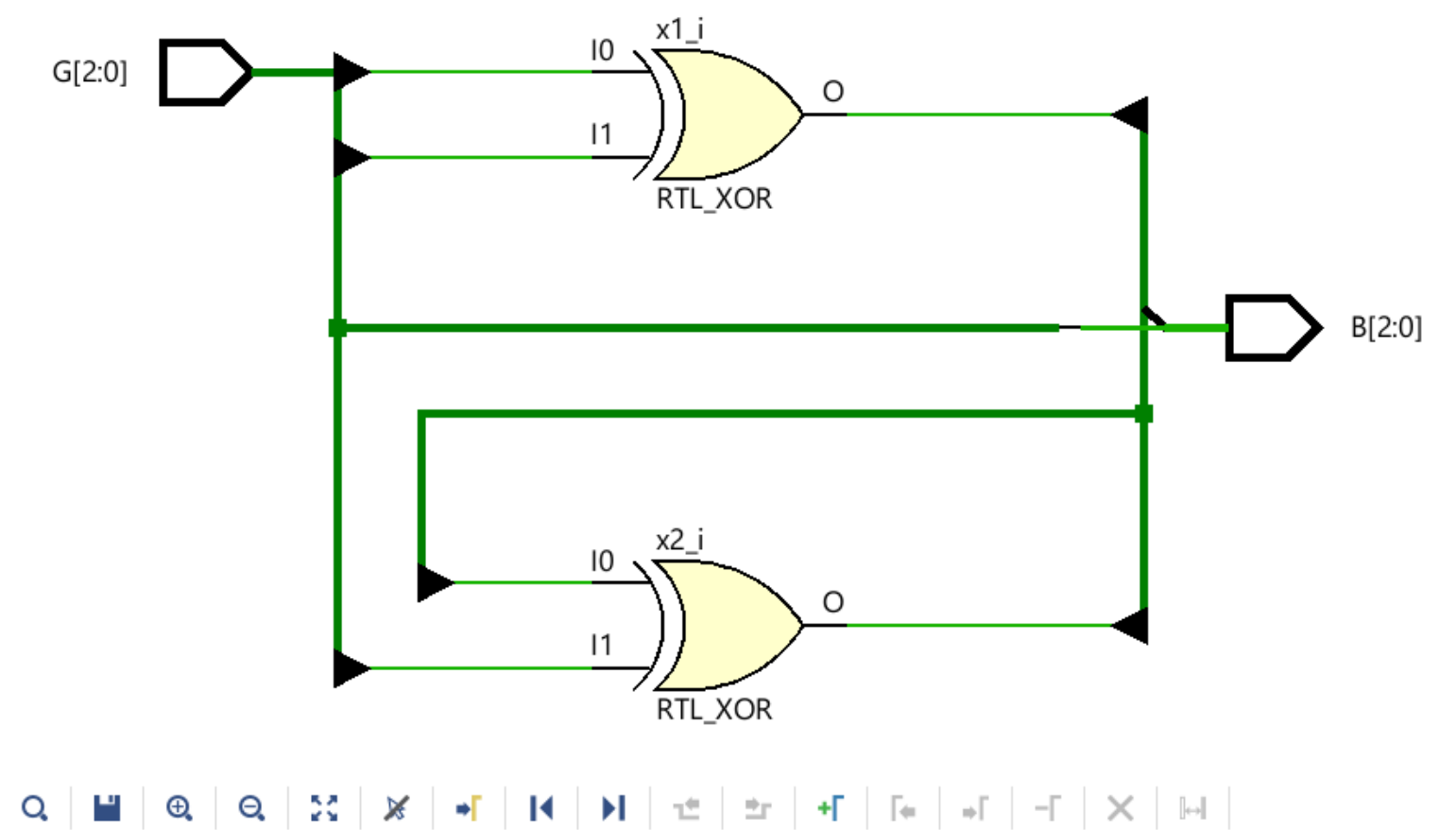
Binary to Gray Code Converter



I2 | I1 | I0 || Y2 | Y1 | Y0

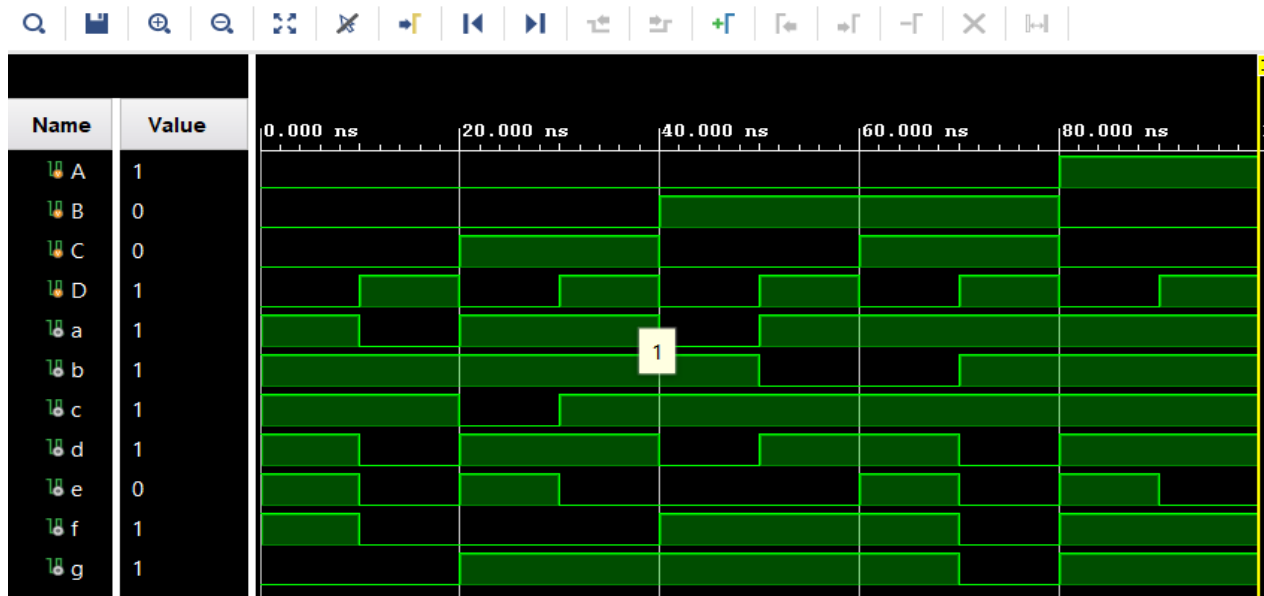
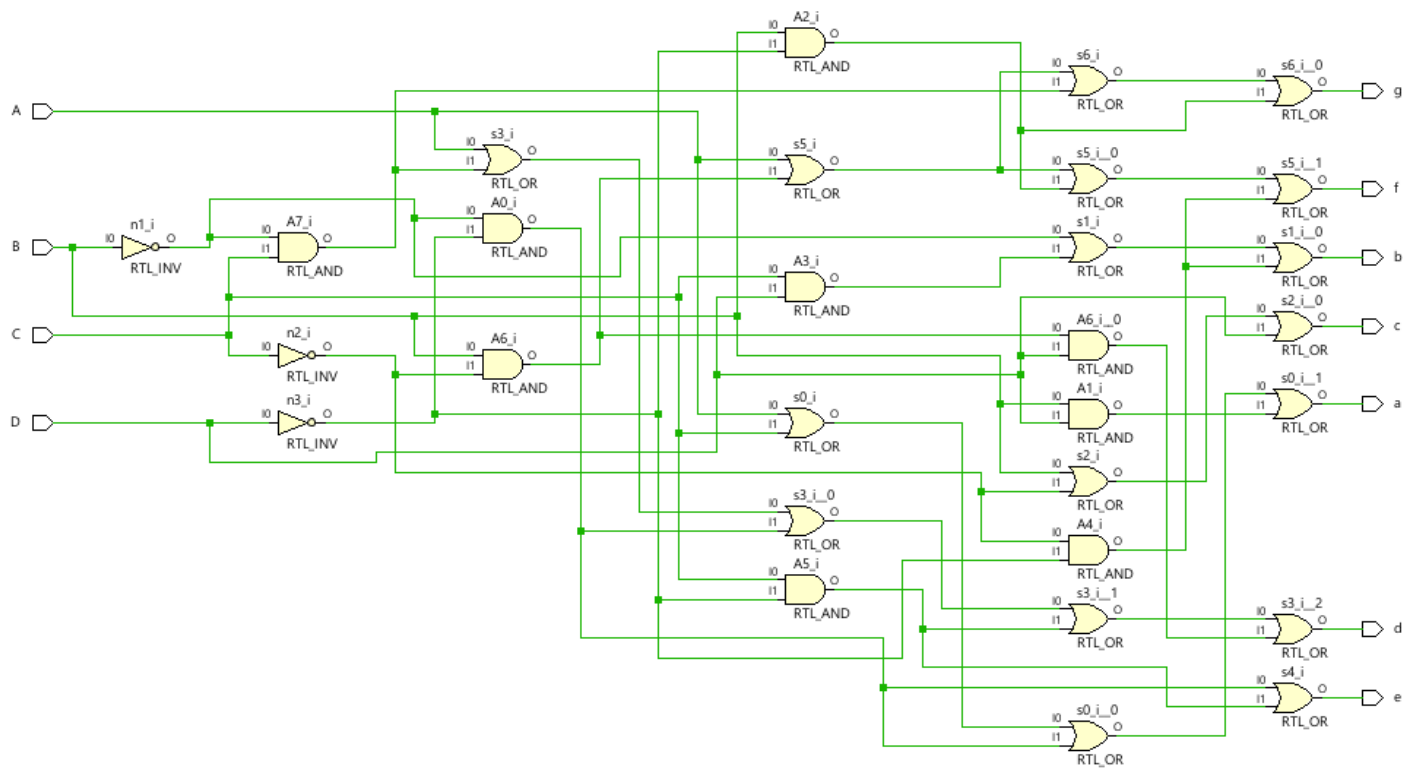
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	1	0	0

Gray to Binary Converter



I2	I1	I0	Y2	Y1	Y0
0	0	0	0	0	0
0	0	1	0	0	1
0	1	1	0	1	0
0	1	0	0	1	1
1	1	0	1	0	0
1	1	1	1	0	1
1	0	1	1	1	0
1	0	0	1	1	1

Binary to 7 Segment



A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1

Behavioral Modeling

